BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SAHIWAL

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: 4191

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics - Paper-II - Objective

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 4191

Note: Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question. No credit will be awarded in case BUBBLES are not filled. Do not solve question on this sheet of OBJECTIVE PAPER.

Q1. If $f(x) = \sqrt{(x+4)}$ then $f(x^2+4)$ is equal to:

(A) $\sqrt{(x^2+8)}$ (B) $\sqrt{(x^2-8)}$

(C) $\sqrt{(x-8)}$ (D) x^2-8

Q2. The function $f(x) = (2+3x)/(2x)$ is not continuous at:

(A) $x = -3$ (B) $x = -2/3$

(C) $x = 1$ (D) $x = 0$

Q3. $\lim(x\to a)\ (f(x)-f(a))/(x-a) =$ ____

(A) f'(x) (B) f'(a)

(C) f'(0) (D) f'(x-a)

Q4. f(x) = x^(2/3), then f'(8) = ____

(A) 1/2 (B) 2/3

(C) 1/3 (D) 3

Q5. The derivative of $(x^3+2x^2)/x^3$ equals:

(A) $2/x^2$ (B) $-2/x^2$

(C) $1/(2x^2)$ (D) $-1/(2x^2)$

Q6. If $f(x) = \tan^{-1}x$, then $f'(\cot x)$ is equal to:

(A) $1/(1+x^2)$ (B) $\sin^2x$

(C) $\cos^2x$ (D) $\sec^2x$

Q7. $f(x+\delta x)$ = ____

(A) $f'(x)\delta x$ (B) $f(x) - f'(x)\delta x$

(C) $f(x) + f'(x)\delta x$ (D) $f(x)\delta x$

Q8. $\int(a/(x\sqrt{(x^2-1)}))dx$ =

(A) $a \tan^{-1}x$ (B) $-a \operatorname{cosec}^{-1}x + c$

(C) $-a \sec^{-1}x + c$ (D) $1/a \sec^{-1}x + c$

Q9. When the expression $\sqrt{(a^2-x^2)}$ involves in integration substitute, is:

(A) $x = a \sin \theta$ (B) $a \sec \theta$

(C) $a \tan \theta$ (D) $a = \sin \theta$

Q10. $\int_0^3 1/\sqrt{(9-x^2)}\, dx$ = ____

(A) $2/\pi$ (B) $-2/\pi$

(C) $-\pi/2$ (D) $\pi/2$

Q11. Which of the following is not a solution of the system of inequalities $x+2y \leq 8$, $2x-3y \leq 6$, $2x+y \geq 2$, $x \geq 0$, $y \geq 0$

(A) (1,0) (B) (8,0)

(C) (0,4) (D) (3,0)

Q12. When axes are translated, the coordinates of the point (-6,9) are changed into (-3,7), find the point through which axes are translated:

(A) (-3,2) (B) (3,-2)

(C) (7,-3) (D) (-9,6)

Q13. The equation of horizontal line passing through (-5,3) is:

(A) $x+5=0$ (B) $-5x+3y=0$

(C) $3x-5y=0$ (D) $y-3=0$

Q14. A line passes through (1,5) and (k,7) and has a slope k, the values of k is:

(A) -1 and 2 (B) 3 and -2

(C) 2, -3 (D) -1, -2

Q15. The focus of the parabola $y^2 = 4ax$ is:

(A) (a,0) (B) (0,a)

(C) (-a,0) (D) [illegible in source scan]

Q16. The eccentricity of $y^2/4 - x^2 = 1$ equals:

(A) √2/√5 (B) 2/√5

(C) -√5/2 (D) √5/2

Q17. Conic are the curves obtained by cutting a right circular cone by:

(A) Sphere (B) A line

(C) A plane (D) A curve

Q18. If a and b are two non-zero vectors, then a×b = ____

(A) b×a (B) a·b

(C) -a×-b (D) b×a [best-effort]

Q19. Angle between the vectors l+j, l-j is:

(A) π (B) π/2

(C) π/4 (D) 0

Q20. Projection of a along b is:

(A) $\hat{a}\cdot\hat{b}$ (B) a-b

(C) $a\cdot\hat{b}$ (D) $\hat{a}\cdot b$

Mathematics - Paper-II - Subjective

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: NOTE: Write same question number and its part number on answer book, as given in the question paper.

Note: SECTION-I

Q2.. (Marks 16) Attempt any eight parts. 8 x 2 = 16

(i) Determine whether the function f(x) = sin x + cos x is even or odd.

(ii) With out finding the inverse, state domain and range of f^{-1} where f(x) = (x-1)/(x-4), x ≠ 4

(iii) Evaluate the limit Lim(x→2) $(x^3-8)/(x^2+x-6)$ by using algebraic techniques.

(iv) Express the Limit Lim(x→0) (1+2x²)^(1/x²) in terms of e.

(v) Find the derivative of (x+4)^(1/3) by definition.

(vi) Differentiate $x^3 - 1/x^2$ w.r.t x^4.

(vii) If $y = \ln(x+\sqrt{(x^2+1)})$ then find dy/dx

(viii) If $y = x^2 \cdot e^{-x}$ then find y_2

(ix) If $x = 1-t^2$ and $y = 3t^2-2t^3$ then find dx/dt and dy/dt

(x) If $f(x) = 4-x^2$, $x \in (-2,2)$ then find interval in which f(x) is increasing or decreasing.

(xi) Prove that $d/dx(\tan^{-1}x) = 1/(1+x^2)$

(xii) If $y = \sin 3x$ then find y_4

Q3.. (Marks 16) Attempt any eight parts. 8 x 2 = 16

(i) Using differentials find dy/dx and dx/dy if $x^2+2y^2 = 16$

(ii) Evaluate $\int \cos 3x \sin 2x \, dx$

(iii) Evaluate $\int (x^2/(4+x^2)) \, dx$

(iv) Evaluate $\int x \ln x \, dx$

(v) Evaluate $\int ((a-b)x/((x-a)(x-b))) \, dx$, $a > b$

(vi) Evaluate $\int_0^3 dx/(x^2+9)$

(vii) Solve the differential equation $ydx + xdy = 0$

(viii) Evaluate $\int \sec x \, dx$

(ix) Find K so that the line joining A(7,3), B(K,-6) and the line joining C(-4,5), D(-6,4) are parallel.

(x) Find whether the given point R(5,8) lies above or below the line $2x-3y+6=0$

(xi) Determine value of P such that the lines $2x-3y-1=0$, $3x-y-5=0$ and $3x+Py+8=0$ meet at a point.

(xii) Find the lines represented by $3x^2+7xy+2y^2=0$

> Note: (2)

Q4.. (Marks 18) Attempt any nine parts. 9 x 2 = 18

(i) Graph the solution set of linear inequality in xy-plane $3x-2y \geq 6$

(ii) Find the equation of a circle with ends of a diameter at (-3,2) and (5,-6)

(iii) Find the centre and radius of a circle $4x^2+4y^2-8x+12y-25=0$

(iv) Write down the equation of normal to the circle $x^2+y^2=25$ at (4,3)

(v) Find the vertex and directrix of $x^2 = 4(y-1)$

(vi) Write the equation of parabola with focus (-3,1) and directrix $x-2y+3=0$

(vii) Find the equation of hyperbola with Foci (±5,0) and vertex is (3,0)

(viii) Find the magnitude of vector u = i+j

(ix) Find a unit vector in the direction of v = i+2j-k

(x) Find the direction cosines of v = 3i-j+2k

(xi) If u = [2,-3,1], v = [2,4,1] find the cosine of angle θ between u and v

(xii) If a×b=0 and a·b=0, what conclusion can be drawn about a or b?

(xiii) Find α so that i-j+k, i-2j-3k and 3i-αj+5k are coplanar.

> Note: SECTION-II

Q5.. (a) If f(x) = { (√(2x+5)-√(x+7))/(x-2), x ≠ 2; K, x = 2 } Find the value of K so that f is continuous at x = 2 (b) If x = a $\cos^3\theta$, y = b $\sin^3\theta$, show that a(dy/dx) + b tan θ = 0

Q6.. (a) Evaluate the integral ∫(√2/(sin x + cos x)) dx (b) Find the condition that the lines $y = m_1x + c_1$, $y = m_2x + c_2$, $y = m_3x + c_3$ are concurrent.

Q7.. (a) Evaluate $\int_0^{\pi}$ (sin x - 1)/$\cos^2x$ dx (b) Maximize f(x,y) = 2x+5y subject to constraints 2y-x ≤ 8, x-y ≤ 4, x ≥ 0, y ≥ 0

Q8.. (a) Write equations of two tangents from (2,3) to the circle $x^2+y^2=9$ (b) By using vectors prove that cos(α+β) = cosα cosβ - sinα sinβ

Q9.. (a) If $y = (\cos^{-1}x)^2$, prove that $(1-x^2)y_2 - xy_1 - 2 = 0$ (b) Show that an equation of parabola with focus at (a cosα, a sinα) and directrix x cosα + y sinα + a = 0 is $(x \sin\alpha - y \cos\alpha)^2 = 4a(x \cos\alpha + y \sin\alpha)$

GROUP – II

Mathematics - Paper-II - Objective

Time Allowed: 30 Minutes	**Maximum Marks:** 20	**Paper Code:** 4192

> Note: Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that bubble in front of that question number, on bubble sheet. Use marker or pen to fill the bubbles. Cutting or filling two or more bubbles will result in zero mark in that question. No credit will be awarded in case BUBBLES are not filled. Do not solve question on this sheet of OBJECTIVE PAPER.

Q1. The perimeter P of a square as a function of its area A is:

(A) $P = \sqrt{A}$ (B) $P = 2\sqrt{A}$

(C) $P = 3\sqrt{A}$ (D) $P = 4\sqrt{A}$

Q2. $\lim_{x\to 3} (x-3)/\sqrt{(x-\sqrt{3})} =$ ____

(A) $\sqrt{3}$ (B) $2\sqrt{3}$

(C) $1/\sqrt{3}$ (D) $1/(2\sqrt{3})$

Q3. If $3x+4y-5=0$, then $dy/dx =$

(A) 4/3 (B) -4/3

(C) 3/4 (D) -3/4

Q4. $d/dx(\sqrt{(\cot x)}) =$

(A) $1/(2\sqrt{(\cot x)})$ (B) $\operatorname{cosec}^2 x/(2\sqrt{(\cot x)})$

(C) $-\operatorname{cosec}^2 x/(2\sqrt{(\cot x)})$ (D) $2\operatorname{cosec}^2 x/\sqrt{(\cot x)}$

Q5. If $f(x) = \tan^{-1} x$, then $f'(\cot x) =$

(A) $\sin^2 x$ (B) $\cos^2 x$

(C) $\sec^2 x$ (D) $1/(1+x^2)$

Q6. $d/dx(-\cot x) =$

(A) $\sec^2 x$ (B) $\operatorname{cosec}^2 x$

(C) $-\operatorname{cosec}^2 x$ (D) $\tan^2 x$

Q7. $\int a^x dx =$

(A) $a^x + c$ (B) $a^x + \ln a + c$

(C) $a^x \cdot \ln a + c$ (D) $a^x/(\ln a) + c$

Q8. The anti-derivative of $1/((1+x^2)\tan^{-1} x)$ is:

(A) $\ln(\tan^{-1} x) + c$ (B) $\ln(1+x^2) + c$

(C) $2(\tan^{-1} x)^2 + c$ (D) $1/2(\tan^{-1} x)^2 + c$

Q9. Suitable substitution for solving $\int 1/(x\sqrt{(x^2-a^2)})\, dx$ is:

(A) $x = a \sin \theta$ (B) $x = a \tan \theta$

(C) $x = a \sec \theta$ (D) $x = a \cos \theta$

Q10. $\int_0^{(\pi/4)} \sec^2 x\, dx =$

(A) 0 (B) 1

(C) 2 (D) 1/2

Q11. If (3,5) is the mid-point of (5,a) and (1,7), then a = ____

(A) 3 (B) 5

(C) 7 (D) 9

Q12. The point (3,-8) lies in the ____ quadrant.

(A) 1st (B) 2nd

(C) 3rd (D) 4th

Q13. The lines l_1 and l_2 with slopes m_1 and m_2 respectively, are parallel if:

(A) $m_1m_2 = 1$ (B) $m_1 = m_2$

(C) $m_1m_2 = -1$ (D) $m_1 + m_2 = 0$

Q14. The point (2,1) is not in the solution of the inequality:

(A) $2x+y > 3$ (B) $2x+y > 4$

(C) $2x+y < 3$ (D) $2x+y > 1$

Q15. Centre of the circle $x^2+y^2+7x-3y = 0$, is:

(A) (7,-3) (B) (-7/2,3/2)

(C) (-7,3) (D) (7/2,-3/2)

Q16. The equation of directrix of the parabola $x^2 = 5y$ is:

(A) x+5/4=0 (B) x-5/4=0

(C) y+5/4=0 (D) y-5/4=0

Q17. The length of latus-rectum of hyperbola $x^2/a^2 - y^2/b^2 = 1$, is:

(A) $a^2/(2b)$ (B) $b^2/(2a)$

(C) $2b^2/a$ [best-effort] (D) $2b^2/a$ [best-effort]

Q18. If $u = 2\alpha i + j - k$ and $v = i + \alpha j + 4k$ are perpendicular, then α =

(A) -4/3 (B) 4/3

(C) 3/4 (D) 4

Q19. The vectors u, v and w are coplanar if:

(A) $u \cdot v \times w = 0$ (B) $u \cdot v \times w = 1$

(C) $u \cdot v \times w = 2$ (D) $u \cdot v \times w = 3$

Q20. Work done by a constant force $\bar{F}$ during a displacement $\bar{d}$ is equal to:

(A) $\bar{F} \times \bar{d}$ (B) $\bar{d} \times \bar{F}$

(C) $\bar{F} + \bar{d}$ (D) $\bar{F} \cdot \bar{d}$

Mathematics - Paper-II - Subjective

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: NOTE: Write same question number and its part number on answer book, as given in the question paper.

Note: SECTION-I

Q2.. (Marks 16) Attempt any eight parts. 8 x 2 = 16

(i) Express perimeter P of a square as a function of its area A.

(ii) If f(x) = √(x+1), g(x) = $1/x^2$ find fog(x), gof(x)

(iii) Evaluate Lim(x→-1) $(x^3-x)/(x+1)$

(iv) Evaluate Lim(x→0) $(1-\cos x)/\sin^2 x$

(v) Differentiate $(\sqrt{x} - 1/\sqrt{x})^2$ w.r.t. x

(vi) Find dy/dx if $x = 1-t^2$, $y = 3t^2-2t^3$

(vii) Prove that $d/dx(\tan^{-1}x) = 1/(1+x^2)$

(viii) Find dy/dx if y = x cos y

(ix) Find dy/dx if y = x/lnx

(x) Find f'(x), if f(x) = e^(√x+1) [best-effort]

(xi) Find y_2 if $y = x^2e^{-x}$

(xii) Find Maclaurin series for sin x.

Q3.. (Marks 16) Attempt any eight parts. 8 x 2 = 16

(i) Use differentials to find dy and δy if $y = x^2+2x$, x changes from 2 to 1.8.

(ii) Find ∫((1-√x)/√x) dx

(iii) Find $\int((1-x^2)/(1+x^2))\,dx$

(iv) Find ∫(cot√x/√x) dx

(v) Find ∫(√2/(sin x + cos x)) dx

(vi) Find ∫x lnx dx

(vii) Solve the differential equation $(e^x+e^{-x})dy/dx = e^x-e^{-x}$

(viii) Find $\int_0$^(π/2) $\cos^2\theta \sin\theta\, d\theta$

(ix) By means of slope, show that (-1,-3), (1,5), (2,9) lie on the same line.

(x) Check whether the point (-7,6) lies above or below the line 4x+3y-9=0

(xi) Check whether the lines 12x+35y-7=0 and 105x-36y+11=0 are parallel or perpendicular.

(xii) Express 15y-8x+3=0 in normal form.

> Note: (2)

Q4.. (Marks 18) Attempt any nine parts. 9 x 2 = 18

(i) Graph the solution set of $3x-2y \geq 6$

(ii) Find an equation of the circle with ends of a diameter at (-3,2) and (5,-6)

(iii) Find the radius of the circle $5x^2+5y^2+14x+12y-10=0$

(iv) Find the length of the tangent drawn from the point (-5,4) to the circle $5x^2+5y^2-10x+15y-131=0$

(v) Find the focus and directrix of the parabola $x^2 = 4(y-1)$

(vi) Find the foci and eccentricity of $y^2/16 - x^2/9 = 1$

(vii) Write down the equation of tangent to $3x^2+3y^2+5x-13y+2=0$ at (1,10/3)

(viii) Find the magnitude of the vector u = i+j

(ix) Find a unit vector in the direction of v = 2i-j

(x) Let v = 3i-2j+2k, w = 5i-j+3k find v-3w

(xi) Find α so that $|\alpha i+(\alpha+1)j+2k| = 3$

(xii) Find the direction cosines of v = 3i-j+2k

(xiii) Find a vector of lengths 5 in the direction opposite that of v = i-2j+3k

> Note: SECTION-II

Q5.. (a) Prove that $y(dy/dx) + x = 0$ if $x = (1-t^2)/(1+t^2)$, $y = 2t/(1+t^2)$ (b) If $f(x) = \{ (\sqrt{(2x+5)}-\sqrt{(x+7)})/(x-2), x \neq 2; K, x = 2 \}$ find value of K so that f is continuous at x = 2

Q6.. (a) Determine value of p such that the lines 2x-3y-1=0, 3x-y-5=0 and 3x+py+8=0 meet at a point. (b) Evaluate $\int x^3 \cdot 5^x \, dx$ [best-effort]

Q7.. (a) Evaluate $\int_{-1}^{2} (x+|x|) \, dx$ (b) Minimize z = 2x+y; subject to the constraints $x+y \geq 3$; $7x+5y \leq 35$; $x \geq 0, y \geq 0$

Q8.. (a) Show that the circles $x^2+y^2+2x-2y-7=0$ and $x^2+y^2-6x+4y+9=0$ touch externally. (b) A force of magnitude 6 units acting parallel to 2i-2j+k, displaces, the point of application from (1,2,3) to (5,3,7). Find work done.

Q9.. (a) A box with a square base and open top is to have a volume of 4 cubic dm. Find the dimensions of the box which will require the least material. (b) Find the centre, foci and vertices of the following $9x^2-12x-y^2-2y+2=0$